
Global CO₂ Emissions

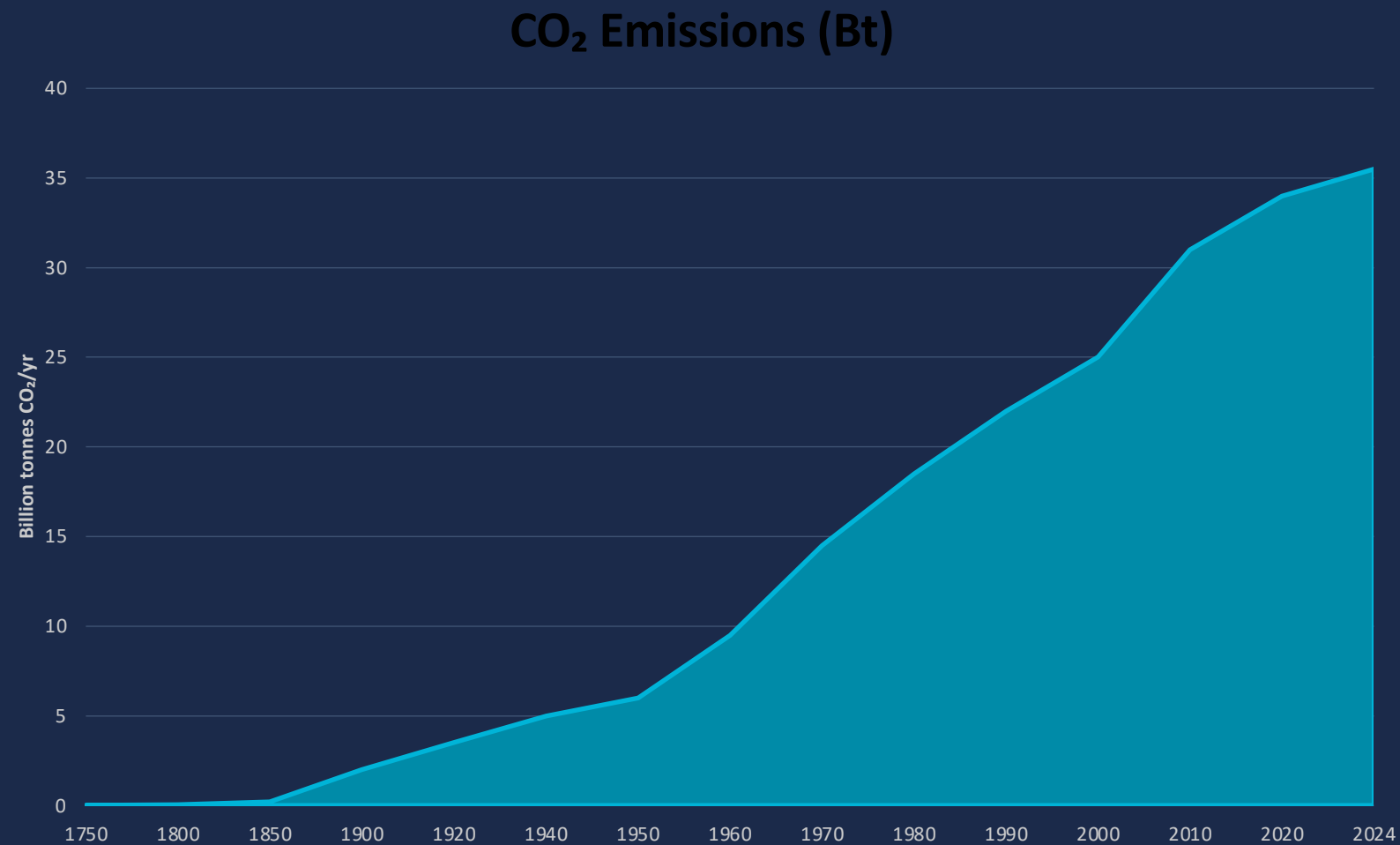
Who Emits, How Much, and What's Changing

Fossil Fuels & Industry | 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

Audience: Policy analysts, climate negotiators, sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



- Pre-1900: Negligible emissions
- 1950: ~6 Bt/yr — post-war industrialization begins
- 1990: ~22 Bt — nearly quadrupled in 40 years
- 2024: >35 Bt — growth slowing, but no peak yet

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

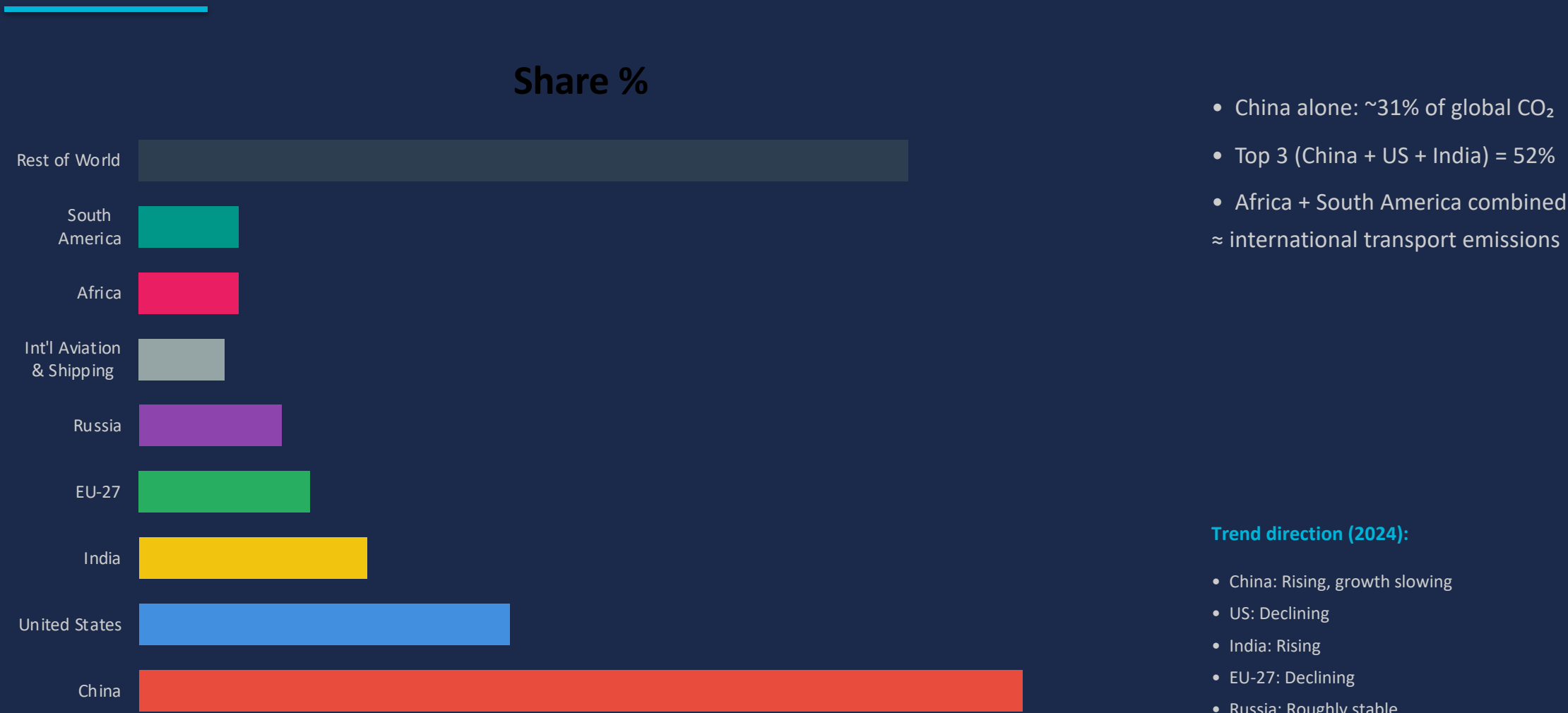


STACKED AREA CHART — Regional Emission Shares Over Time

Y-axis: % of global CO₂ emissions | X-axis: 1850–2024
Regions: Europe, North America, Asia, Rest of World
Shows Europe+US declining from >90% (1900) to <33% (2024)
Asia (led by China) rising to ~50% of global total

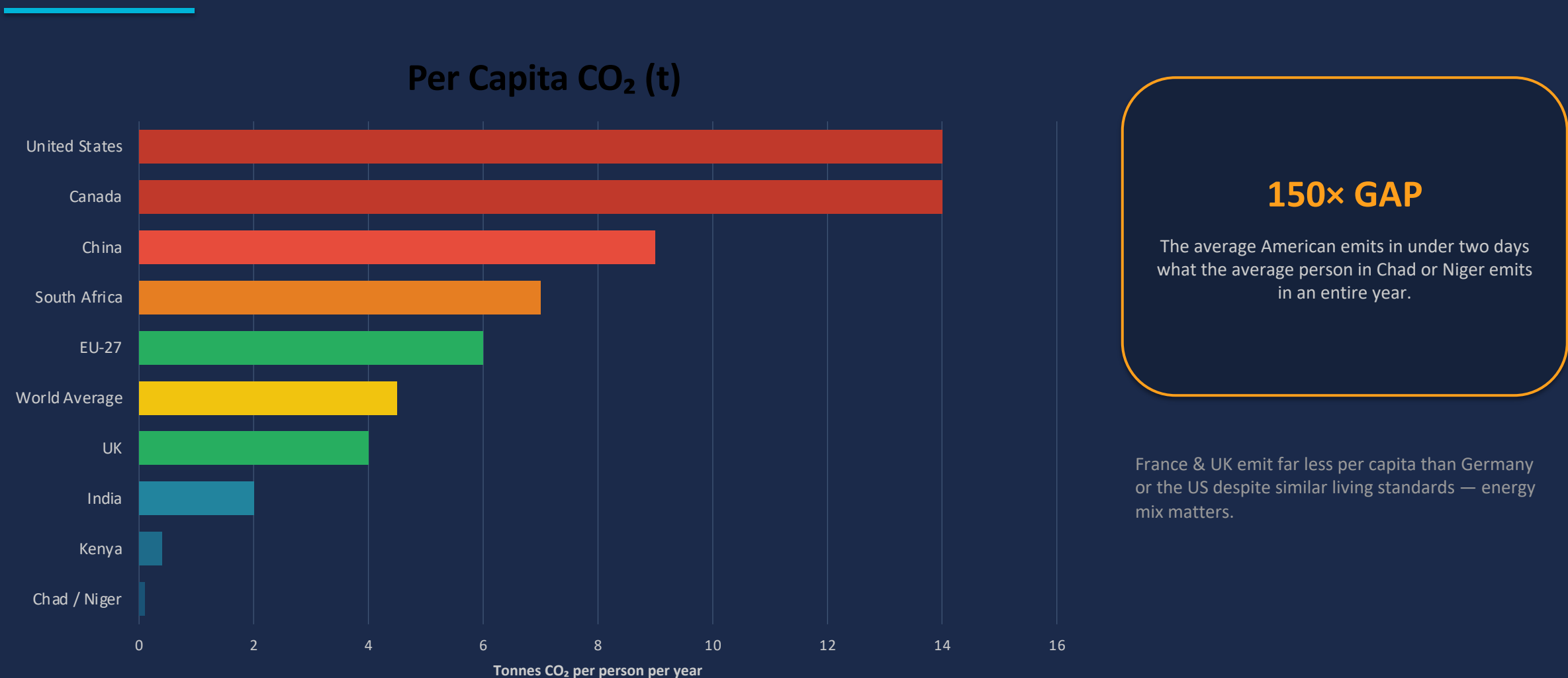
- 1900: Europe + US > 90%
- 1950: Europe + US still ~85%
- Today: Europe + US < one-third
- Asia now produces ~50% of global emissions

China Now Emits More Than One-Quarter of Global CO₂



Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions: ~14 t in the US vs. ~9 t in China and ~2 t in India

MULTI-LINE CHART — Per Capita CO₂ Trends by Country

Y-axis: tonnes CO₂ per person per year | X-axis: 1950–2024
Lines: US, Canada, China, EU-27, UK, India, World avg
US/Canada steady at ~14 t; China rose from <1 t to ~9 t;
India remains low at ~2 t; EU declining toward ~6 t

2024 Snapshot (per capita):

United States	~14 t
Canada	~14 t
China	~9 t
South Africa	~7 t
EU-27	~6 t
World average	~4.5 t
United Kingdom	~4 t
India	~2 t
Kenya	~0.4 t

China's per capita is still below the US despite being the largest total emitter.

2024: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



DIVERGING BAR CHART — Annual % Change in CO₂ Emissions, 2024

Center axis: 0% change | Left (blue): declining | Right (orange/red): increasing

Declining: US, EU, UK — shown in blue

Increasing: Russia, Southeast Asia, parts of Africa — shown in orange/red

Central Asia (e.g. Turkmenistan/Uzbekistan): >10% increase

DECLINING

US, EU, UK and much of Western Europe

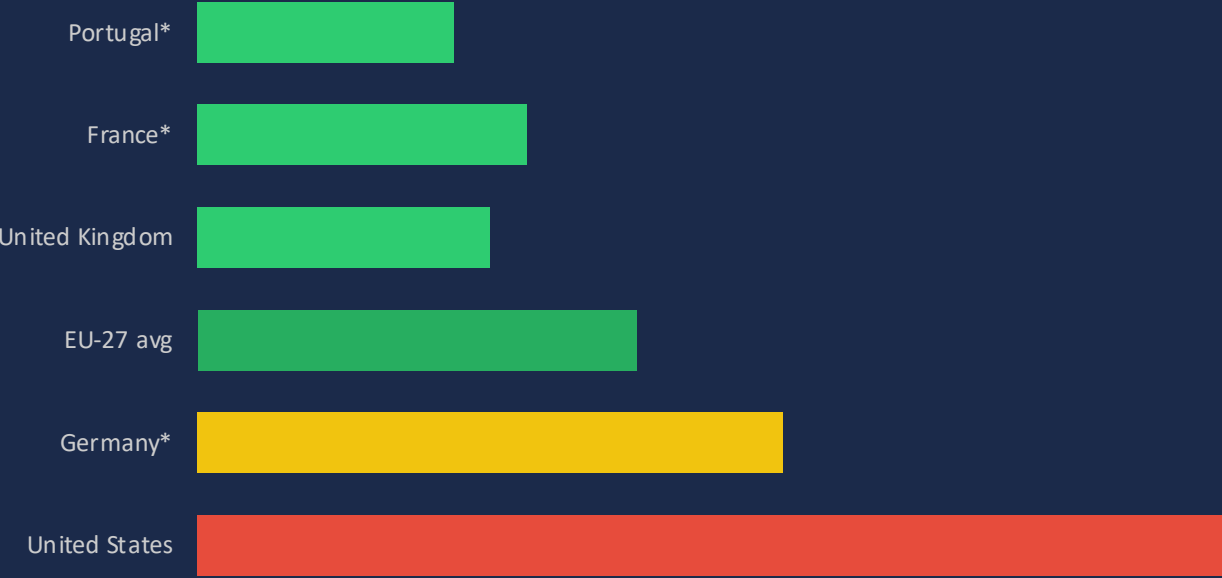
INCREASING

Russia, Southeast Asia, several African nations

Central Asia (e.g. Turkmenistan/Uzbekistan) saw >10% increases

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Per Capita CO₂ (t)



* Approximate values based on EU-27 avg of ~6 t and known relative positions

WHY THE DIFFERENCE?

Nuclear & renewables

France, UK, and Portugal generate a much higher share of electricity from nuclear and renewable sources.

Fossil fuel reliance

Germany, the Netherlands, and the US still rely heavily on coal and natural gas for electricity.

Key takeaway

Similar prosperity levels do not require similar emission levels — energy policy choices drive the difference.

Key Takeaways: Responsibility Is Shared but Unequal

1

No peak yet

Global CO₂ emissions from fossil fuels exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.

2

China leads annual emissions, but context matters

China is the largest annual emitter (~31%), but its per capita emissions (~9 t) remain below the US (~14 t). Historical responsibility is dominated by Europe and the US.

3

Historical responsibility is concentrated

Europe and the US accounted for >90% of emissions in 1900. Their cumulative contribution to atmospheric CO₂ far exceeds their current share.

4

Massive per capita inequality persists

A 150× gap exists between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and the highest large-country emitters (US/Canada, ~14 t/person).

5

Energy policy can decouple prosperity from emissions

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix — France and the UK emit far less per capita than Germany or the US.